Joel Williams

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Portfolio QR Code

Education

Georgia Institute of Technology, Atlanta, GA August 2023 – May 2027 (Expected)

Bachelor of Mechanical Engineering, Minor in Computer Science, GPA: 3.95

- **Clubs:** HyTech Racing, NSBE, ASME, ColorStack, Ramblin' Rocket Club, Invention Studio

Experience

Prototyping Instructor, Invention Studio at Georgia Tech, Atlanta, GA November 2023 - Present

- Instruct 30+ general users per week on manufacturing methods, DFM practices, and safe equipment use.
- Coach prospective prototyping instructors in safe machine operation, standard operating procedures, and teaching methods for general studio users, achieving a ~90% pass rate for those trained and tested.
- Operate Tormach and EMCO CNCs to produce precision prototype parts; utilize Autodesk Fusion to create CAM toolpaths, select tooling, and tune feeds/speeds across jobs.
- Support general users' CNC jobs end-to-end by assisting with setup, DFM considerations, fixturing, and process planning to translate CAD designs into manufacturable parts.

Reliability Engineering Intern, Valero, Houston, TX May 2025 – August 2025

- Built operating cost (OC) models for steam turbines and presented conversion cases to reduce OC by 85%.
- Conducted failure mode and effects analysis (FMEA) on a bad-actor pump; implemented structural fixes raising MTBR over 340% compared to the previous baseline and authored the investigation report.
- Optimized select pump operating parameters across 5 refinery units to reduce critical machine downtime; added real-time dashboard indicators to make deviations from optimal operation visible and actionable.
- Led access-structure upgrades; scoped catwalk, platform, and walkway installs, coordinated 3 contractor site visits, and consolidated bids, improving mechanic safety and decreasing maintenance route times by 50%.

Research Assistant, Kennesaw State University, Kennesaw, GA May 2022 – May 2023

Synthesis of conductive polymer nanomaterials to develop more efficient and cost-effective supercapacitors.

- Evaluated the capacitance and cycling stability of synthesized nanomaterials with CV and GCD testing.
- Utilized SEM and XRD to analyze nanomaterial morphology and composition, guiding purity optimization and maximizing charge capacity for future test samples.
- Collaborated with a multidisciplinary team to compile and synthesize graphical data representations for conferences, symposium events, and reports.

Projects

Focus Puck | Smart Desktop Productivity Device: Atlanta, GA December 2025 – December 2025

- Defined user requirements, BOM, and an intuitive assembly process for an interactive desktop focus timer with touch controls, LED light feedback, and audio cues to support structured study sessions.
- Applied DFAM principles by optimizing wall thickness, minimizing overhangs, and adding clearances.
- Incorporated modular features including wire routing, locating pegs, and headers, enabling flexible iteration.
- Verified durability with ANSYS stiffness/load checks and 20+ drop tests from desk height (~0.75 m).

GNC | Thrust-Vector-Control Mount: Atlanta, GA August 2024 – March 2025

- Designed a new thrust-vector-control (TVC) mount to improve rocket active control capability, improving gimbal range of motion by ~400% on both pitch and yaw axes compared to the previous design.
- Selected materials and processes for component manufacturing: MSLA resin for low-friction joint interface, PLA-CF for high-heat applications, and PETG for moderate heat and impact resistance.
- Tested and revised TVC servo linkages to reduce mechanical play and improve reliability across launches.
- Sourced cost-effective COTS hardware to enable rapid iteration with field-replaceable components.

GNC | Fin-Tabs-Actuated Full Rocket: Atlanta, GA August 2025 - Present

- Calculated snatch forces and bolt shear, ensuring a FoS of 3, and cross-validated with ANSYS simulations.
- Integrated rocket subsystems into a top-down SolidWorks assembly, providing a single reference for geometry, interfaces, and cross-team integration decisions.
- Developed a detailed 2-month manufacturing timeline, aligning sub-team tasks to ensure all components met completion deadlines for the planned launch date.
- Led manufacturing tasks including waterjetting and milling bulkheads, and 3D printing avionics bays, applying GD&T principles to achieve first-pass fit and reliable subsystem integration.

Technical Skills & Coursework

Technical Skills: SolidWorks (CSWA Certified), CREO, Windchill, ANSYS, Fusion, CATIA, GD&T, DFMA, FMEA, CNC/Mill

Relevant Coursework: Mechanics of Materials, Deformable Bodies, Thermodynamics, Fluid Mechanics, Dynamics of Rigid Bodies, Creative Decisions and Design, Statics, Heat Transfer, System Dynamics